

Nomura Quant Challenge 5.0

Python Environment Setup Guide

Step 1: Install Anaconda

Download Anaconda:

- Go to the Anaconda Distribution page: <https://www.anaconda.com/download>
- Download the installer for your operating system (Windows, macOS, or Linux).

Install Anaconda:

- **Windows:**
 - Double-click the downloaded .exe file and follow the installation instructions.
- **macOS:**
 - Open the downloaded .pkg file and follow the installation instructions.
- **Linux:**
 - Open a terminal and navigate to the directory where you downloaded the installer.
 - Run the following command (replace filename with the correct one):

```
bash Anaconda3-2023.11-Linux-x86_64.sh
```
 - Follow the prompts to complete the installation.

Verify Installation:

Open a terminal or Anaconda Prompt (Windows) and type:

```
conda --version
```

You should see the version of conda printed on the screen, confirming the installation.

Step 2: Create a Conda Environment

Create the Environment:

1. Download and save the environment_base.yml file provided with this guide.
2. Open a terminal or Anaconda Prompt (Windows).

3. Navigate to the directory where your environment_base.yml file is located.
4. Run the following command:

```
conda env create -f environment_base.yml
```

Activate the Environment:

Once the environment is created, activate it with:

```
conda activate nqc3
```

Verify the Environment:

Check the installed packages and Python version by running:

```
conda list
```

environment_base.yml Contents:

```
name: nqc3
channels:
- defaults
- conda-forge
dependencies:
- python=3.13
- numpy
- pandas
- scipy
- matplotlib
- seaborn
- scikit-learn
- statsmodels
- jupyterlab
- xlrd
- openpyxl
- notebook
```

Step 3: Using the Conda Environment with IDEs

Jupyter Notebook / JupyterLab:

5. Navigate to your working folder in the command prompt.
6. Activate your environment: `conda activate nqc3`
7. Start JupyterLab: `jupyter lab`
8. Alternatively, start Jupyter Notebook: `jupyter notebook`

PyCharm:

9. Download and install PyCharm from <https://www.jetbrains.com/pycharm/download/>
10. Open PyCharm and create or open your project.

11. Go to File > Settings (or PyCharm > Preferences on macOS).
12. Navigate to Project: <project_name> > Python Interpreter.
13. Click the gear icon and select Add.
14. Choose Conda Environment and select Existing environment.
15. Point to the Python executable in your nqc3 conda environment:

Linux/macOS: `.../anaconda3/envs/nqc3/bin/python`

Windows: `.../Anaconda3/envs/nqc3/python.exe`

VS Code:

16. Download and install VS Code from <https://code.visualstudio.com/>
17. Press Cmd+Shift+P (Ctrl+Shift+P on Windows).
18. Search for "Select Interpreter".
19. Select it and choose the nqc3 environment from the list.

You are free to use other IDEs as well (they should support conda environments).

Submission Requirements

IMPORTANT: Participants must submit a **.py file** as their solution.

- Your submission must be a Python script (.py file) that can be executed in the nqc3 conda environment specified above.
- Ensure your code runs without errors in the provided environment (nqc3 with Python 3.13).
- Do not use any packages outside of those listed in the `environment_base.yml` file.
- Submit your solution as a .zip file containing:
 - Your Python solution file (.py)
 - A .docx file explaining your approach
 - Any trained model in pickle format (if required)

Please ensure that submission guidelines are followed properly and the code follows the virtual environment specified to prevent entry disqualification.

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